**Question:**

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

A. 2.48 × 10^-8 cm

B. 5.92 × 10^7 cm/sec

C. 8.66 × 10^-6 cm

D. 3.14 × 10^-5 cm

E. 9.81 × 10^-4 cm

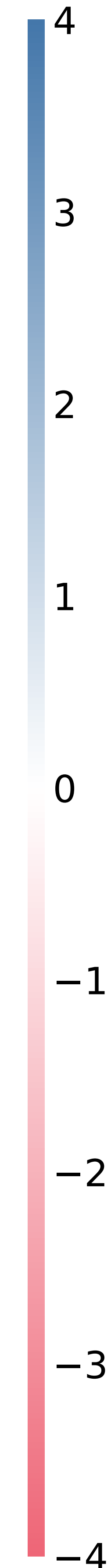
F. 5.27 × 10^-9 m

G. 1.23 × 10^-7 cm

H. 4.14 × 10^-10 m

I. 6.625 × 10^-27 erg sec

J. 0.512 × 10^6eV




**Reasoning + Answer:**

<think>

de Broglie wavelength is given by the formula Î» = h / p , where h is Planck's constant and p is the momentum of the particle . The momentum of the electron can be calculated as p = âĪ ļ ( 2 * m * E ), where m is the electron mass , and E is the kinetic energy in electron volts ( e V ).

Since energy in electron volts ( e V ) can be converted to joules ( J ) using the conversion factor 1 e V = 1 . 6 ÃĹ 1 0 ^- 1 9 J , the kinetic energy in joules is 1 . 6 ÃĹ 1 0 ^- 1 9 J * 1 e V / 1 . 6 ÃĹ 1 0 ^- 1 9 J .

The momentum can then be calculated as âĪ ļ ( 2 * 9 . 1 ÃĹ 1 0 ^- 3 1 kg * 1 . 6 ÃĹ 1 0 ^- 1 9 J ).

Planck's constant h is approximately 6 . 6 2 5 ÃĹ 1 0 ^- 3 4 J Â· s .

Substituting the momentum and Planck's constant into the formula Î» = h /p , the wavelength in meters ( m ) can be calculated and then converted to centimeters ( cm ) by multiplying by 1 0 0 .

</think>

<answer>

F . 5 . 2 7 ÃĹ 1 0 ^- 9 m

</answer>